Raunak Vernekar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.27
S.E.S.Junior College Solapur	Class XII (Maharashtra State Board)	2023	84.50%
B.F. Damani Highschool	Class X (Maharashtra State Board)	2021	99.60%

SKILLS

- **Programming**: C, C++, Python, JavaScript, SQL
- **CS Fundamentals:** DSA, OOP, DBMS, OS, Computer Networks
- **Web & Backend:** React.js, Node.js, Express.js, FastAPI, REST APIs, MongoDB, PostgreSQL, Redis, pgvector
- **Machine Learning:** scikit-learn, Regression, Classification, Feature Engineering, Model Evaluation
- **Deep Learning & Computer Vision:** PyTorch, TensorFlow, YOLO, ByteTrack
- **Generative AI:** LLMs, RAG, Temporal RAG, Semantic Search
- **Tools & Systems:** Linux, Bash, Git, GitHub, CMake, Ninja, CTest, Postman

RELEVANT COURSES

- **Programming:** Introduction to Programming
- **Maths:** Introduction to Probability, Differential Equations, Calculus, Linear algebra, Complex Variables
- **AI:** Introduction to Modern AI.

PROJECTS

SentinelAI - AI-Powered Video Intelligence & Temporal RAG

- Built a long-video analysis pipeline using **YOLO, ByteTrack, and adaptive frame sampling** for object tracking and timestamped event detection.
- Developed **Temporal RAG with PostgreSQL/pgvector** and LLM-based Q&A for semantic, timestamp-grounded video querying and scene summarization.
- Built an **asynchronous FastAPI/Celery/Redis backend** and React dashboard for video processing, event timelines, and interactive querying.
- **Tech Stack:** Python, FastAPI, Celery, Redis, YOLO, ByteTrack, PostgreSQL, pgvector, React, Groq

Comparative Study of Logistic Regression (Scratch vs scikit-learn) - Titanic Survival Prediction:

- Built an end-to-end ML pipeline to predict passenger survival using the Titanic dataset.
- Implemented **Logistic Regression from scratch** with NumPy and compared it against **scikit-learn Logistic Regression, Decision Tree, Random Forest, and XGBoost** models
- Preprocessed data (imputation, scaling, encoding) using ColumnTransformer; evaluated with **k-Fold cross-validation**.
- Analyzed performance via **ROC curves, loss trends, and accuracy metrics** for learning efficiency and generalization.
- **Tech Stack:** Python, NumPy, Pandas, scikit-learn, XGBoost, Matplotlib, Seaborn.

IIT ResourceHub - Full-Stack Campus Utility Platform (MERN Stack)

- Developed a **full-stack campus platform** integrating **Q&A Forum, academic resources, clubs, events, Lost & Found, bookmarks, and academic planning**.
- Implemented **JWT-based authentication with user/admin role-based access control and authorization middleware** for protected REST APIs.
- Built **MongoDB-backed modules, CRUD operations, file uploads, and responsive React/Tailwind interfaces** for seamless student interaction.
- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, JWT, Multer, Tailwind CSS.

MiniGit - Git-Inspired Version Control System | C++20

- Built a **Git-inspired version-control system from scratch** with repository initialization, staging, commits, branches, checkout, status, log, and line-level diffing.
- Implemented **SHA-256 content-addressed storage** with Blob, Tree, and Commit objects, persistent staging index, and **parent-linked commit history**.
- Designed a modular filesystem-backed architecture with **automated regression testing and CTest**, covering hashing, commits, history, branching, checkout, and diff workflows.
- **Tech Stack:** C++20, CMake, Ninja, STL, std::filesystem, Bash, CTest

POSITIONS OF RESPONSIBILITY

- **Tinkerers' Lab, IITH**- Mechatronics Core: 2025-26 (academic year)

Actively involved in hands-on projects related to Mechatronics and ROS as part of the lab's core team.

- **EML (Extra Mural Lectures) Club, IITH**- Design Coordinator: 2025-26 (academic year)

Design social media creatives, posters, and visual content for lectures conducted by renowned industry speakers.

EXTRACURRICULAR

- **Athletics:** Competitive cricket and badminton
- **Interests:** Realistic portrait drawing and structured fitness training